

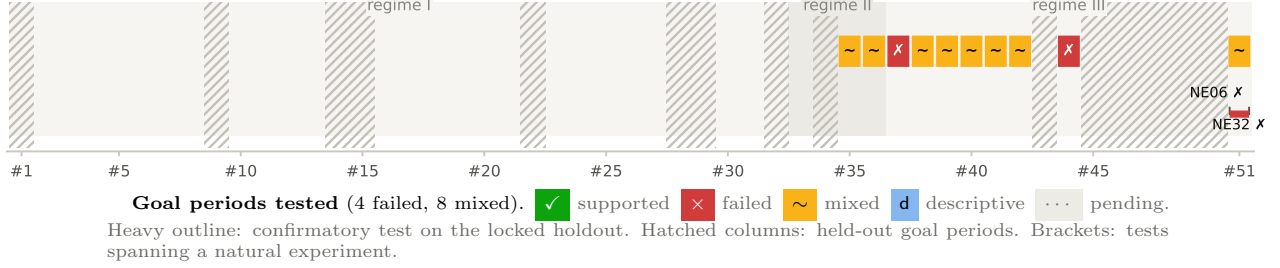
# H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2/5

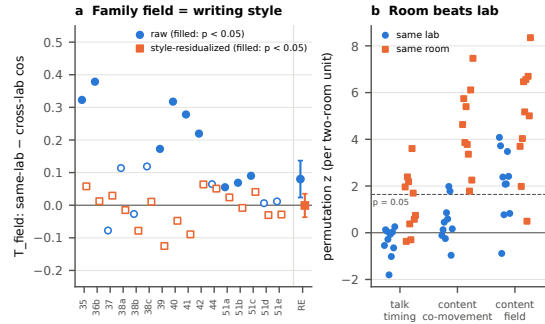
**Bottom line.** Model families differ in how they write and only faintly in what they do, not in what they think or whom they follow; rooms, not vendors, set who couples with whom.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

15 counted units (#35–#44, #51; non-holdout;  $N = 10\text{--}27$ ). After removing each day’s common field, same-lab agents share a content direction: 9/15 units, random-effects  $T = 0.08$  [0.02, 0.14], stable across periods (split-half 0.88 vs 0.76,  $p = 0.008$ ). **It is writing style:** it survives style residualization in 0/9 units. **No family coupling** in talk timing or content co-movement; **rooms carry coupling.** **Round 1b** (second embedding model, DQ5 style residuals, dedupe, corrected talk table) reproduces every number; the talk null is 2.3× tighter. The new behavioral test (Jev v3 states, real failures, commit and write rates) **failed** its rule: 3/15 units, RE  $T_B = 0.075$  [0.03, 0.12], a weak work-habit trace. Natives: NE06 (Google-only change) is village-wide in the logs; newcomers 3/9 by behavior ( $p = 0.14$ ); #35 mixed. Holdout not run.



**Observables.** (a) Family-field statistic  $T_{\text{field}}$  per unit before (blue) and after (orange) regressing out writing-style features; filled = lab-permutation  $p < 0.05$ ; “RE” = random-effects summary with 95% CI. The field vanishes under style control. (b) Two-room units: permutation  $z$  of the same-lab and same-room pair coefficients. Talk timing and content co-movement follow rooms, not labs; the static content field has a lab term too, which is style.

## PHYSICAL MODEL

Model 11 (vector spins) plus models 01/02 in  $K$ -block mean field.  $v_{i,d}$ : agent  $i$ ’s mean whitened chat embedding on day  $d$ ;  $g_d$ : day (goal) field, estimated by the day mean  $m_d$ ;  $h_f$ : field of lab  $f(i)$ ;  $a_i$ : agent offset;  $x_{i,d}$ : fluctuations, coupled through a  $K \times K$  family matrix ( $K = 2\text{--}5$  multi-member labs);  $\varepsilon$ : sampling noise.  $H_i$ : agent field.  $C$ : talk-spin or co-movement correlation matrix.  $y_{ij}$ : a pair outcome;  $r(i)$ : agent’s room.

$$v_{i,d} = g_d + h_{f(i)} + a_i + x_{i,d} + \varepsilon_{i,d}, \quad H_i = \langle v_{i,d} - m_d \rangle_d$$

$$T_{\text{field}} = \langle \cos(H_i, H_j) \rangle_{f(i)=f(j)} - \langle \cos(H_i, H_j) \rangle_{f(i) \neq f(j)}$$

$$J \approx -C^{-1} \text{ (off-diag.)}, \quad J_{ij} = J_{f(i)f(j)}, \quad \Delta = J_{\text{in}} - J_{\text{out}}$$

$$y_{ij} = c + b_{\text{lab}}[f(i) = f(j)] + b_{\text{room}}[r(i) = r(j)]$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For anyone mixing vendors in one swarm: models from the same lab sound alike, and that is nearly all there is. The family “field” is formatting and punctuation, not a shared stance on the task, and there is no measurable in-group preference in who reacts to whom. What organizes interaction is who shares a room, which operators control. Two practical consequences, both modest. Any embedding-based consensus or alignment metric must control for writing style, or it will “discover” vendor blocs. Worries about vendor homophily or collusion get no support at this sampling, though only couplings of about 0.1 or larger are excluded. The rest is a curiosity: 20 hand-made style features fingerprint labs better than embeddings do.

H13-family-fields · generated 2026-10-04 10:55 by in-fra/summaries/build\_summaries.py · card hypotheses/H13-family-fields/README.md · ratings: RE-C2 round-1b agent (suggested; faithfulness and usefulness unchanged from the coordinator’s calibration)

## RESULTS VS PREDICTIONS

| Prediction                  | Round 1 → round 1b (bge; gte)                       | Verdict   |
|-----------------------------|-----------------------------------------------------|-----------|
| P1 family field             | 9/15, RE 0.080 → 9/15, 0.080; gte mixed 8/15, 0.085 |           |
| P2 not only style           | 0/9 → 0/9 (own S-a); 1/9, 0/8 (DQ5 × residuals)     |           |
| P3 family invariance        | 0.88 vs 0.76 → 0.88 / 0.90 vs 0.76 / ✓ 0.74         |           |
| P5 no talk homophily        | RE 0.002 [−0.04, 0.05] → −0.003 ✓ [−0.02, 0.02]     |           |
| P6 no content homophily     | 1/15 → 1/15 (both models)                           | ✓         |
| P7 rooms carry coupling     | talk 8 → 10/10; co-movement 9/10, ✓ gte 10/10       |           |
| B1 behavioral field (HH267) | 3/15; RE $T_B$ 0.075 [0.03, 0.12]                   | ×         |
| B2 behavior beyond style    | 4/15; RE 0.038 [−0.01, 0.09]                        | ×         |
| NE06 / NE32 / G35 (native)  | village-wide change / 3 of 9 / words follow rooms   | × / × / ~ |

## GOAL PERIODS IN DETAIL

G35 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G36 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G37 × 1b: failed (round 1: failed; same in both embedding m...  
 G38 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G39 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G40 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G41 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G42 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 G44 × 1b: failed (round 1: failed; same in both embedding m...  
 G51 ~ 1b: mixed (round 1: mixed; same in both embedding mod...  
 +2 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 2 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 0 |
| D unfitted predictions | 1 | I transfer        | 1 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

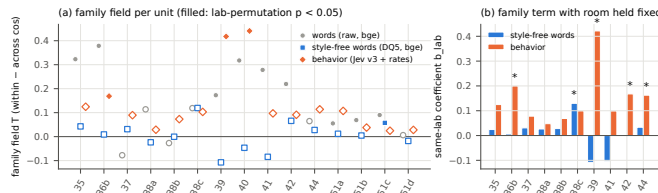
## CAVEATS

The behavioral field may be harness rather than disposition: it lives in file-write and commit cadence, which follow each provider’s tool conventions, and NE06 could not separate the two because the change hit every agent. Jev’s labels read window evidence that can include an agent’s narration. In #51 roles dominate behavior (same-role pairs dropped). The “no coupling” results exclude only couplings  $\gtrsim 0.1$ . Holdout (C1–C6) untouched.

## ROUND 2

Test the behavioral field against a harness covariate (tool-call conventions per provider); use NE20 (Anthropic one call per turn, holdout) only if the confirmatory run is approved.

- **Essence**. Do vendors differ in decisions, not prose?
- **H13-R1** Policy fingerprint: P(next behavior state — current state, context features) differs by family beyond agent identity.
- **H13-R2** Families differ in susceptibility: who listens to whom (HH98).
- **H13-R3** Mixed-vendor periods out-produce single-vendor ones (descriptive, confounded; work ledger).



**Second figure.** (a) Family field per unit in three channels: raw words (gray), words after DQ5 style residualization (blue) and behavior (orange); filled =  $p < 0.05$ . Style-free words carry nothing; behavior carries a little, mostly in the build weeks #39–#40. (b) Two-room units: the same-lab coefficient with room held fixed; behavior keeps a family term (RE 0.14), style-free words do not.